

PRACTICE TEST FOR TEST 3 (STAT-461/561)

- (1) A CBS News poll conducted in the US in November 16-19, 2012 asked “Are you in favor of legalizing the use of marijuana”. In the nationwide poll of $n = 1100$ adults, 517 said that they favored legalization. Write down a 95% confidence interval for the true, unknown population proportion of adults supporting the legalization of marijuana in the US. Next, write down a 98% confidence interval for the same parameter.
- (2) For each combination of sample size and sample proportion given below, find the margin of error (i.e. half-length) for a 99% confidence interval: (a) $n = 400$ and 80 out of 400 said ‘yes’, (b) $n = 1000$ and 355 out of 1000 were in favor of a government proposal, (c) $n = 650$ and 520 out of 650 said that they prefer stricter rules for COVID protection.
- (3) Volunteers who had developed a common cold within the previous 24 hours were randomized to either take zinc lozenges or placebo lozenges every 2 to 3 hours until their cold symptoms were gone. Twentyfive participants took zinc lozenges and twentythree took the placebo. The mean overall duration of symptoms for those who took zinc was 4.5 days with a standard deviation of 1.6 days. For the placebo group, the mean overall duration of symptoms was 8.1 days with a standard deviation of 1.8 days. Assuming that the duration of symptoms distributions for both populations (zinc lozenges and placebo lozenges) are Normal, (a) What is a 95% confidence interval for the true (unknown) population mean overall duration of symptoms for the zinc group? (b) What is a 95% confidence interval for the true (unknown) population mean overall duration for the placebo group? (c) Based on these two intervals, do you have reasons to believe that the true, unknown population means for the two groups are different? Explain.
- (4) It has been long believed that eating oatmeal regularly lowers blood cholesterol. To test this claim, a scientist with the FDA measured the change in blood cholesterol levels of a randomly chosen sample of 25 people in a certain age-group (who ate oatmeal for breakfast for a month) and found a sample-average reduction of 15 points with a standard deviation of 6 points. Is it believable at a 5% significance level that the population-mean reduction in all people of that age-group is **more than 10 points**?
- (5) A political party claims that its presidential candidate has **more than 55%** support in the New England states. To test this claim, a political fact-checking agency took a random sample of 1225 voting-age people from the New England states and found that 685 of them supported that particular candidate. At a 2% alpha level, is the political party’s claim believable? What is the **p-value** of this test?
- (6) How large a sample size do you need to estimate an unknown population proportion by a 99% confidence interval whose half-width is 0.15?
- (7) For a simple random sample of size n from a population, if we decide to use a slightly different version of the sample variance (i.e., with n as the denominator instead of $n-1$) as

a point estimator of the unknown population variance, is it an unbiased estimator? What is its MSE? Is it a consistent estimator?

- (8) For a simple random sample of size n from a Weibull $(2, \lambda)$ population, derive (a) the method-of-moments point estimator for the unknown parameter λ , (b) the maximum likelihood estimator of the unknown parameter λ .
- (9) The lengths of tube worms found around the deep-sea vent near Galapagos Islands are normally distributed with a mean of 4.5 feet and standard deviation 1.4 feet. What are the chances that a randomly selected worm will have length (a) exceeding 6 feet? (b) at most 4 feet? (c) between 4 and 6 feet?
- (10) In the problem above, 95% tube worms are at least how long? What is the 85th percentile of tube-worm lengths? What is the IQR of the tube-worm lengths? [Hint: The IQR is the difference between the 75th and the 25th percentiles]
- (11) If in problem (9), you took a random sample of 49 worms, what are the chances that the sample-average length would be (a) at least 5 feet? (b) at most 3.9 feet? (c) between 4.2 and 4.8 feet? (d) more than the population-mean?
- (12) If X has an Exponential pdf with mean = 10, what is $P(X > 15)$? What is the conditional probability that X exceeds 20, given that X has already exceeded 5?
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- (15) If a discrete random vector (X, Y) has the following joint pmf table, find the marginal pmf of X , its mean and variance. Then find the marginal pmf, mean and variance of Y . Are X and Y independent? If not, compute the correlation between X and Y . Also, what are the conditional pmf of X given that $Y > 5$, $E(X | Y > 5)$ and $\text{Var}(X | Y > 5)$

	$Y=3$	$Y=6$	$Y=9$
$X=0$	0.05	0.08	0.12
$X=1$	0.1	0.16	0.19
$X=2$	0.15	0.1	0.05

